

1 Signals and Systems

** Problem 1.1 (Energy and Power)

Determine which of these signals are energy-type or power-type. In case, find the energy E or power P , respectively.

$$\text{a.) } x(t) = \text{sign } t = \begin{cases} 1 & t \geq 0 \\ -1 & t < 0 \end{cases}$$

Since $|x(t)|^2 = 1$, the signal doesn't have (finite) energy. The power P is given by

$$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{2T}{2T} = 1.$$

$$\text{b.) } x(t) = A \cos(2\pi f_1 t) + B \cos(2\pi f_2 t) \text{ with } f_1 \neq f_2. \quad \text{Hint: } \cos a \cos b = \frac{1}{2} (\cos(a - b) + \cos(a + b))$$

$$\begin{aligned} |x(t)|^2 &= A^2 \cos^2(2\pi f_1 t) + B^2 \cos^2(2\pi f_2 t) + 2AB \cos(2\pi f_1 t) \cos(2\pi f_2 t) \\ &= \underbrace{A^2 \cos^2(2\pi f_1 t)}_{\geq 0} + \underbrace{B^2 \cos^2(2\pi f_2 t)}_{\geq 0} + AB (\cos(2\pi(f_1 - f_2)t) + \cos(2\pi(f_1 + f_2)t)) \end{aligned}$$

Since the first two terms are not-negative, the signal $x(t)$ doesn't have finite energy and is not energy-type.

The power P is given by

$$\begin{aligned} P &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt \\ &= \underbrace{\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T A^2 \cos^2(2\pi f_1 t) dt}_{=\frac{A^2}{2}} + \underbrace{\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T B^2 \cos^2(2\pi f_2 t) dt}_{=\frac{B^2}{2}} \\ &\quad + \underbrace{\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T AB (\cos(2\pi(f_1 - f_2)t) + \cos(2\pi(f_1 + f_2)t)) dt}_{=0, \text{ since } f_1 \neq f_2} \\ &= \frac{A^2 + B^2}{2}. \end{aligned}$$

Mind that if $f_1 = f_2$, the two cosines add coherently and $P = \frac{(A+B)^2}{2}$.

$$\text{c.) } x(t) = 1/t \cdot u(t) \text{ with the unit-step function } u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_0^{\infty} \frac{1}{t^2} dt = \lim_{s \rightarrow 0} -\frac{1}{t} \bigg|_s^{\infty} = \infty$$

Hence, the signal $x(t)$ has infinity energy and is not energy-type.

The same way of reasoning yields that

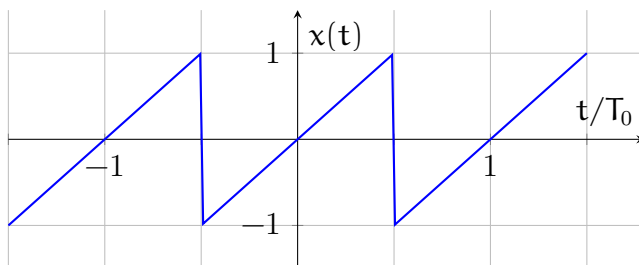
$$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^T \frac{1}{t^2} dt \rightarrow \infty,$$

and the signal $x(t)$ has no finite power.

**** Problem 1.2** (Fourier-Series)

Let a signal $x(t)$ have period T_0 and $x(t) = \frac{2}{T_0} t$ for $|t| < \frac{T_0}{2}$.

a.) Plot the function $x(t)$ for $|t| < \frac{3}{2}T_0$.



b.) Evaluate the Fourier-coefficients X_n of $x(t)$. Hint: $\int t e^{at} dt = \frac{e^{at}}{a^2} (at - 1)$

The Fourier-coefficients X_n are given by

$$X_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) e^{-j \frac{2\pi}{T_0} nt} dt.$$

For $n = 0$ this yields

$$X_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) dt = 0.$$

For $n \neq 0$ and using the provided hint,

$$\begin{aligned} X_n &= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \frac{2}{T_0} t e^{-j\frac{2\pi}{T_0} n t} dt, \quad \text{subst: } u = \frac{2}{T_0} t \\ &= \frac{1}{2} \int_{-1}^1 u e^{-j\pi n u} du \\ &= \frac{1}{2} \frac{e^{-j\pi n u}}{-\pi^2 n^2} (-j\pi n u - 1) \Big|_{-1}^1 \\ &= \frac{j}{2\pi n} (e^{j\pi n} + e^{-j\pi n}) = (-1)^n \frac{j}{\pi n} \end{aligned}$$

c.) Derive the coefficients X_n using the table provided in the lecture notes.

- $x(t)$ is symmetric to the t -axis and hence, the Fourier-coefficient $X_0 = 0$.
- The Fourier-coefficients C_n of the triangular wave given in the notes are

$$C_n = \frac{j}{\pi n}.$$

Mind that the signal raises periodically from $x = -1$ to $x = +1$ and hence, the value $X_0 = 2$ in the table needs to be used (sorry for the overuse of the letter X_0)!

- The signal $x(t)$ is shifted by $\tau = \frac{T_0}{2}$ with respect to the waveform in the notes. By the shift property

$$X_n = e^{-j\frac{2\pi}{T_0} n \tau} C_n = e^{-j\pi n} \frac{j}{\pi n} = (-1)^n \frac{j}{\pi n}.$$

**** Problem 1.3** (Scaling Property)

Let a signal $x(t)$ have period T_0 and consider $y(t) = x(at)$.

a.) Determine the period T of $y(t)$.

If $x(t)$ is periodic with period T_0 , then

$$y\left(t + \frac{T_0}{a}\right) = x\left(at + T_0\right) = x(at) = y(t),$$

and $y(t)$ has period

$$T = \frac{T_0}{a}.$$

b.) Determine the Fourier-coefficients Y_n of $y(t)$ depending on the Fourier-coefficients X_n of $x(t)$.

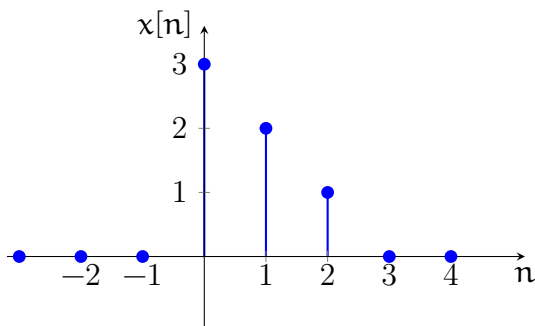
$$\begin{aligned}
 Y_n &= \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t) e^{-j\frac{2\pi}{T}nt} dt \\
 &= \frac{a}{T_0} \int_{-\frac{T_0}{2a}}^{\frac{T_0}{2a}} x(at) e^{-j\frac{2\pi a}{T_0}nt} dt, \quad \text{subst: } t' = at \\
 &= \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t') e^{-j\frac{2\pi}{T_0}nt'} dt' \\
 &= X_n.
 \end{aligned}$$

Hence, the Fourier-coefficients do not change by a scaling of the time-axis. The Fourier-coefficients merely depend on the "shape" of the signal. However, the fundamental frequency changes, which will result in an appropriate stretch of the time-axis.

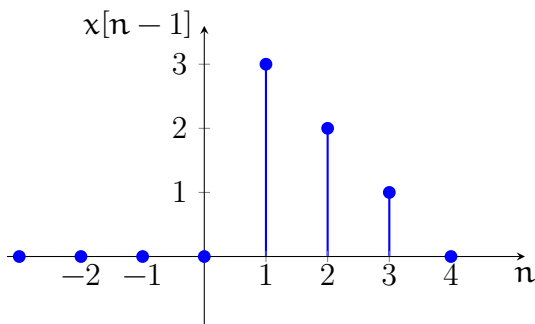
* Problem 1.4 (Discrete-Time Signals)

Let $x[n] = 3\delta[n] + 2\delta[n-1] + \delta[n-2]$. Plot the following signals:

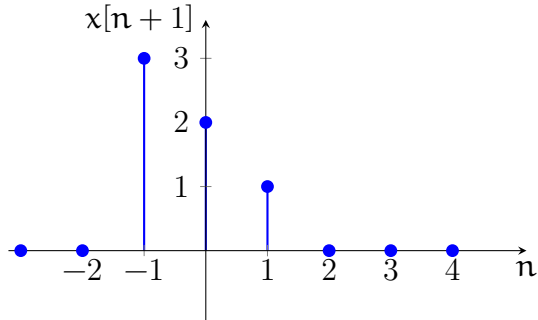
a.) $x[n]$



b.) $x[n-1]$

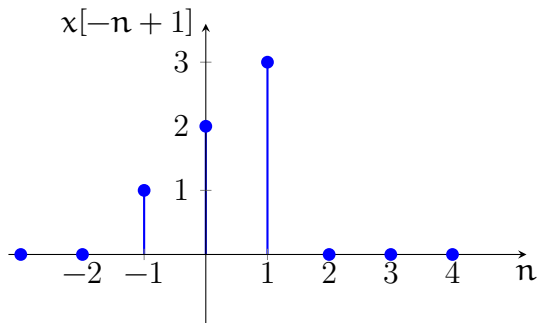


c.) $x[n+1]$

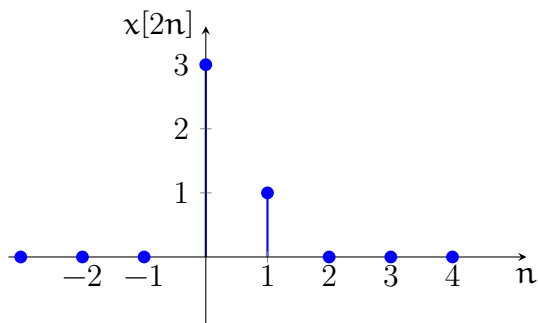


d.) $x[-n+1]$

$$x[-n+1] = x[-(n-1)]$$



e.) $x[2n]$



**** Problem 1.5** (Periodic Signals)

Determine whether each of the following signals is periodic. If the signal is periodic, state its period.

a.) $x[n] = e^{j(\pi n/6)}$

*a.) and b.) are complex exponentials with rational frequency, and hence periodic.
The period can be found by looking at $N_0 \in \mathbb{N}$ that provide integer multiples of 2π .*

$$\pi N_0/6 \equiv 2\pi \quad (= 2\pi k, \text{ for some } k \in \mathbb{Z}) \quad \Rightarrow \quad N_0 = 12$$

b.) $x[n] = e^{j(3\pi n/4)}$

As in part a.), a complex exponential with a rational frequency is periodic.

$$3\pi N_0/4 \equiv 2\pi \quad \Rightarrow \quad N_0 = 8, \quad \text{since } N_0 \text{ must be an integer}$$

c.) $x[n] = \frac{\sin(\pi n/5)}{\pi n}$

The sine-function is periodic, but the denominator πn is not $\Rightarrow x[n]$ is not periodic.

d.) $x[n] = e^{j\pi n/\sqrt{2}}$

The complex exponential has a non-rational number in its argument. Hence, we cannot find any $N_0 \in \mathbb{N}$ such that

$$\pi N_0/\sqrt{2} \equiv 2\pi \quad \text{or} \quad N_0 = 2\sqrt{2}k \quad \text{for some } k \in \mathbb{N},$$

and $x[n]$ is not periodic.

**** Problem 1.6** (System Properties)

Determine whether each of the following systems is memoryless, causal, linear, time-invariant, or stable. $u[n]$ is the unit step signal.

a.) $y[n] = 2x[n] + 1$

The system is memoryless and hence, causal; not (!) linear (it's an affine system, but linearity does not hold); time-invariant; stable.

b.) $y[n+2] = x[n+2] - x[n+1]$

The system can be rewritten as $y[n] = x[n] - x[n-1]$: it has memory, is causal, linear, time-invariant, and stable. The system takes the difference of successive samples and is hence known as a differentiator.

c.) $y[n] = nx[n-1]$

The system has memory due to reference to a past input symbol. The system is causal, linear, but time-variant due to multiplication with time-dependent scaling factor n , and not stable due to the increasing scaling factor n (use the bounded input signal $x[n] = u[n]$ to achieve arbitrarily high output samples).

d.) $y[n] = a^n x[n+1] u[n]$

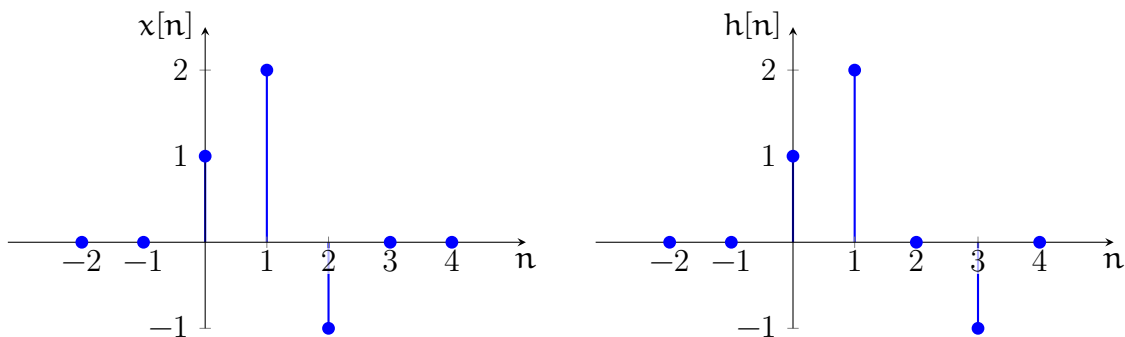
The system has memory and is anticipating of the future, hence non-causal. The system is linear and time-variant due to time-dependent scaling factor a^n as well as the switch $u[n]$. The system is stable only for $|a| \leq 1$.

$$\text{e.) } y[n] = \cos(x[n]) u[n]$$

The system is memoryless and hence, causal, but not linear due to the cosine function. The system is also time-variant due to the multiplication with the switch $u[n]$. The system is stable as the output signal always is bounded.

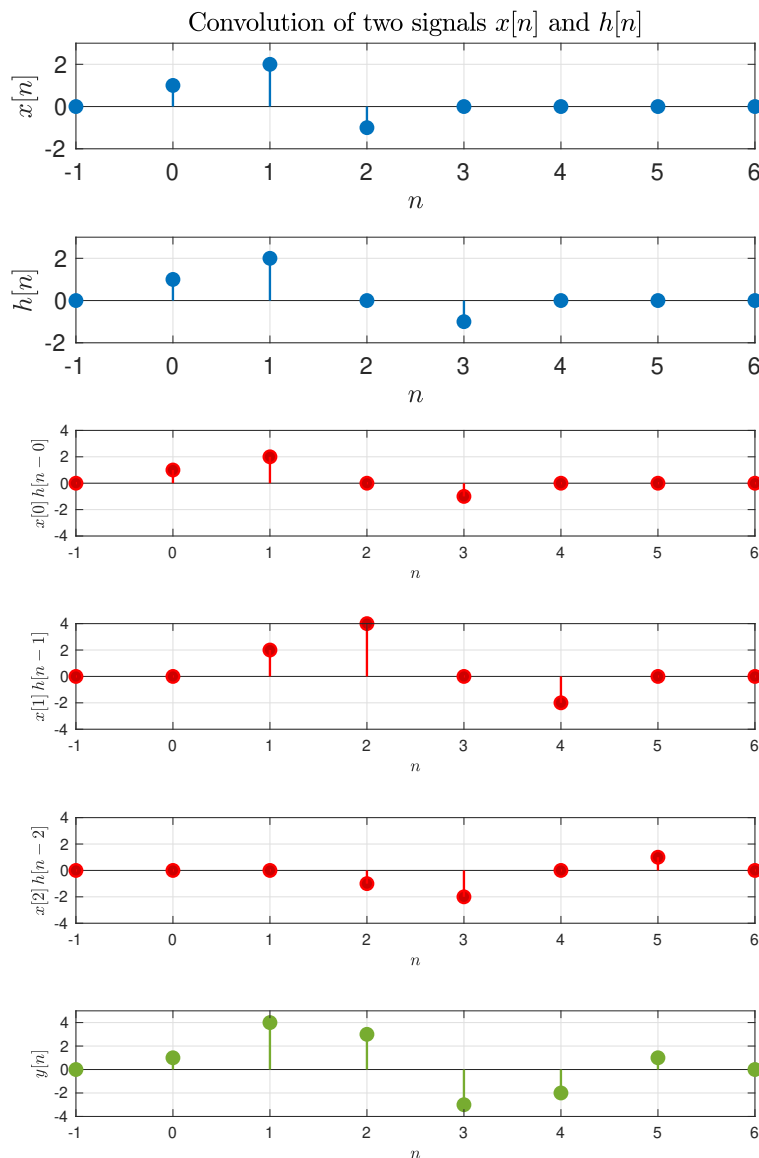
* Problem 1.7 (Convolution)

For the following input signal $x[n]$ and an LTI-system with impulse response $h[n]$, find the output signal $y[n]$.



Using the definition of convolution leads to the direct-sum method of adding scaled and delayed impulse responses. Mind that we apply the fact that the input signal $x[n]$ as well as the impulse response $h[n]$ are causal signals.

$$\begin{aligned} y[n] &= x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = \sum_{k=0}^n x[k] h[n-k] \\ &= x[0] h[n] + x[1] h[n-1] + x[2] h[n-2] \\ &= h[n] + 2 h[n-1] - h[n-2] \end{aligned}$$



Using your *myconv*-MATLAB function to check:

```

» x = [1 2 -1];
» h = [1 2 0 -1];
» y = my_conv(x, h)
y = 1 4 3 -3 -2 1

```

* Problem 1.8 (Duration of Signals)

If $x[n]$ is zero, except for L_x consecutive points, and $h[n]$ is zero, except for L_h consecutive points, what is the maximum number L_y of consecutive points for which $y[n]$ can be non-zero if $y[n] = x[n] * h[n]$?

$x[n]$ has length L_x non-zero samples, and $h[n]$ has length L_h non-zero samples. To have a non-zero

output, the non-zero signal samples and the non-zero impulse response samples need to overlap in at least one position. Hence, the maximum length L_y of the output sequence $y[n]$ is given by

$$L_y = L_x + L_h - 1.$$

**** Problem 1.9** (Closed-Form Convolution)

The impulse response $h[n]$ of an LTI-system is given by $h[n] = a^n u[n]$. Show that the response to the input signal $x[n] = u[n] - u[n - N]$ is given by

$$y[n] = \frac{1 - a^{n+1}}{1 - a} u[n] - \frac{1 - a^{n-N+1}}{1 - a} u[n - N].$$

The geometric sum might be useful:

$$\sum_{k=0}^n a^k = \frac{1 - a^{n+1}}{1 - a}.$$

If $x[n] = u[n]$ as input signal, i.e., an input signal which is 1 for $n \geq 0$ only (otherwise 0):

$$y[0] = x[0] h[0] = h[0] = a^0 = 1$$

$$y[1] = x[1] h[0] + x[0] h[1] = h[0] + h[1] = 1 + a$$

$$y[2] = x[2] h[0] + x[1] h[1] + x[0] h[2] = h[0] + h[1] + h[2] = 1 + a + a^2.$$

Hence, in general

$$y[n] = 1 + a + a^2 \dots + a^n = \frac{1 - a^{n+1}}{1 - a} u[n].$$

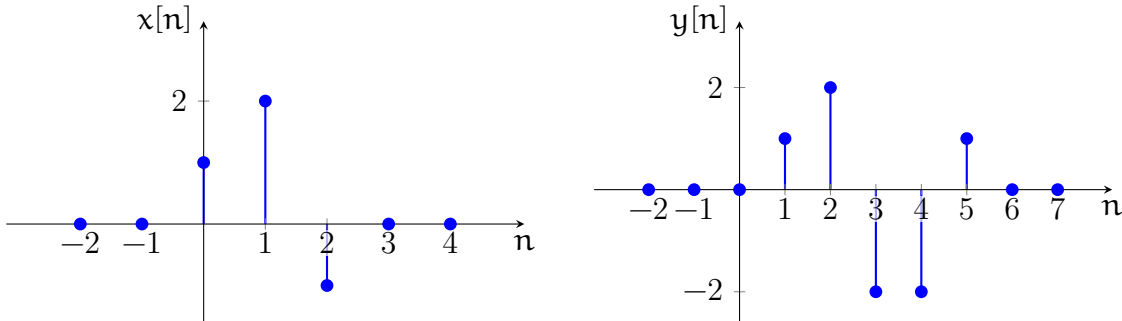
Well, a proof by induction is required from a mathematical point of view – but we skip this part and believe in our engineering gut feeling.

Using that the input signal is a superposition of two time-delayed unit-steps and the linearity property of the system:

$$\begin{aligned} y[n] &= h[n] * x[n] = h[n] * (u[n] - u[n - N]) = \\ &= h[n] * u[n] - h[n] * u[n - N] \\ &= \frac{1 - a^{n+1}}{1 - a} u[n] - \frac{1 - a^{n-N+1}}{1 - a} u[n - N]. \end{aligned}$$

**** Problem 1.10** (De-Convolution)

For an LTI-system, the output $y[n]$ is observed when using the signal $x[n]$ as input. Find the impulse response $h[n]$ of the LTI-system.



The system is causal, since there is no output prior to the first input. Observe that $x[n]$ has 3 consecutive non-zero samples, $y[n]$ contains 5 non-zero consecutive samples. Using Problem 1.8, this yields that the impulse response must have 3 non-zero consecutive samples (in this case, for $1 \leq n \leq 4$):

$$y[n] = x[n] * h[n] = \sum_{k=0}^2 x[k] h[n - k].$$

Processing output sample by output sample:

$$\begin{aligned} y[0] &= x[0]h[0] = h[0] \stackrel{!}{=} 0 \quad \Rightarrow \quad h[0] = 0 \\ y[1] &= x[1]h[0] + x[0]h[1] = h[1] \stackrel{!}{=} 1 \quad \Rightarrow \quad h[1] = 1 \\ y[2] &= x[2]h[0] + x[1]h[1] + x[0]h[2] = 2 + h[2] \stackrel{!}{=} 2 \quad \Rightarrow \quad h[2] = 0 \\ y[3] &= x[3]h[0] + x[2]h[1] + x[1]h[2] + x[0]h[3] = -1 + h[3] \stackrel{!}{=} -2 \quad \Rightarrow \quad h[3] = -1 \\ y[4] &= x[4]h[0] + x[3]h[1] + x[2]h[2] + x[1]h[3] + x[0]h[4] = -2 + h[4] \stackrel{!}{=} -2 \\ &\Rightarrow \quad h[4] = 0. \end{aligned}$$

One might also use MATLAB's `deconv`-function (obviously not available during exams):

```
» x = [1 2 -1];
» y = [0 1 2 -2 -2 1];
» h = deconv(y,x)
h = 0 1 0 -1
```

**** Problem 1.11** (Systems)

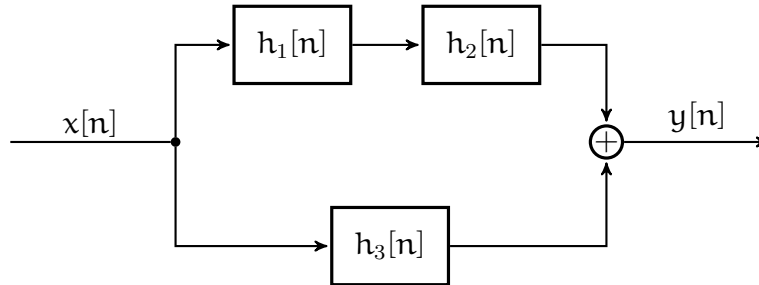
Given the system shown in the figure below with

$$h_1[n] = \delta[n] + \delta[n-1] + \delta[n-2],$$

$$h_2[n] = \delta[n+1] - \delta[n-1],$$

$$h_3[n] = \delta[n-3],$$

find the equivalent impulse response $h[n]$ of the overall system.



Mind that $h_1[n]$ is a series of 3 unit impulses starting at $n = 0$. A serial concatenation of system 1 and 2 yields an LTI-system with impulse response

$$h_4[n] = h_1[n] * h_2[n] = h_1[n] * (\delta[n+1] - \delta[n-1]) = h_1[n+1] - h_1[n-1].$$

The overall impulse response $h[n]$ is the sum of the impulse response $h_4[n]$ of the upper branch and the impulse response $h_3[n]$ of the lower branch:

$$h[n] = h_3[n] + h_4[n] = \delta[n+1] + \delta[n] - \delta[n-2].$$

